IIT Bombay RACING

NRB BEARINGS

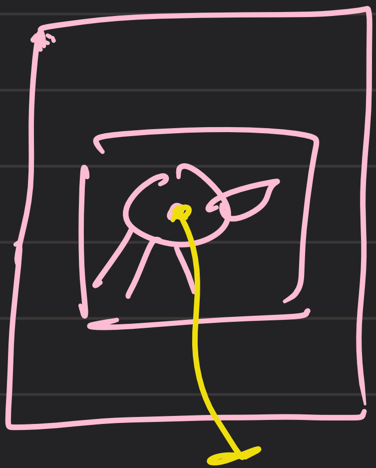
Driverless (perception)

Perception

meet-3

C_1: dog
C_2: person

Yolo : you only look once



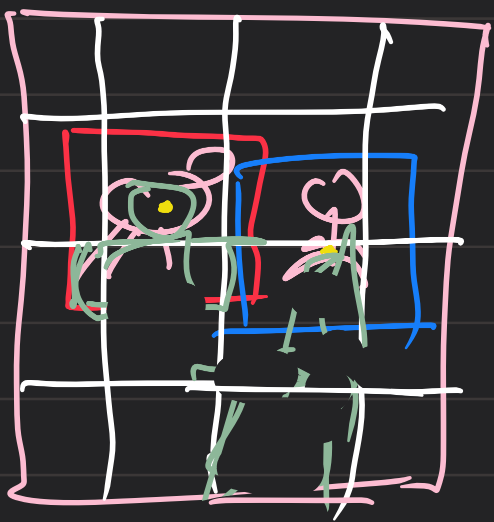
centre point

$$\begin{bmatrix} P_c \\ P_x \\ P_y \\ P_w \\ P_b \\ C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 50 \\ 70 \\ 60 \\ 40 \\ 1 \\ 0 \end{bmatrix}$$

P_c: probability that the image inside the box is either of the object from the class which we have defined

$\in [0,1]$

grids : (4, 8, 16 ---)



→ the $P_c \neq 1$ for all the grids which does not have the centre point

Over lapping :

we will use the method of intersection / union

the value should be greater than the threshold value

↳ min (0.65)

$$\frac{\text{intersection area}}{\text{union area}}$$

↳ max for the one will be choosen

we use 2 cameras
to inc FOV

let say C_1 gives
the image of first
cone at 0.5 sec
& C_2 at 0.5 ± 0.1 sec

yolo takes input
from both

But till
the time C_2 detects
the cone the car has moven

$C_1 : d_1 = 4m$
$C_2 : d_2 = 3.2m$

we define
a radius
& both the
d_1 & d_2 comes
inside it.

HW:
yolo11s.pt

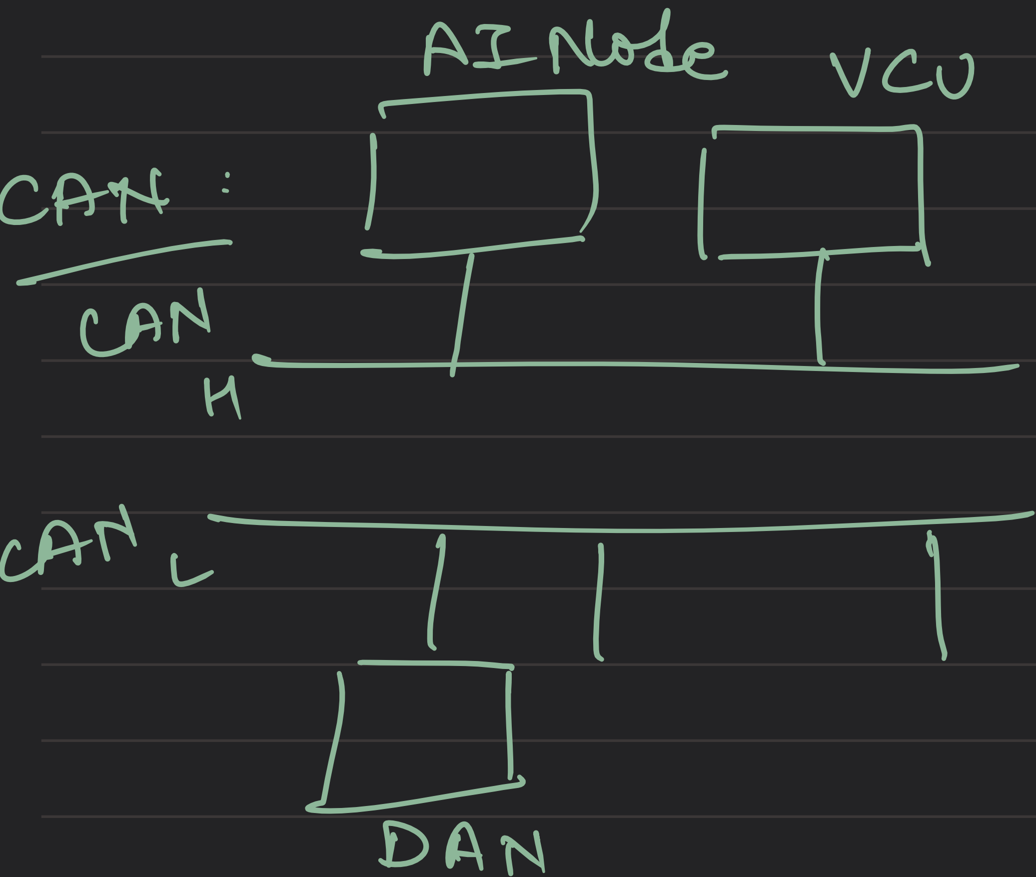
DV → throttle
Steering
Brakes

PCC gives to elee

JETSON

(AI NODE) gives the 3 controls

CAN :



driverless acteuater Node

takes steering & breaks input

There are two modes — manual mode

AS_STATE → S state Autonomous mode

AS_OFF

↓

AS_READY

↓

using ASR

AS_DRIVING DV

AS_FINISHED

AS_EMERGENCY

LIDAR

Meet-4

messages
- point cloud 1 — (x, y, z)
  - Ientensity
- point cloud 2 (16 bit
  - (x, y, z, I)

LIDAR → gives point could

So, what is the 1st thing we do with the point could?

→ remove grou'd points (we don't what groud to hinder our object detection)

How to remove ground?

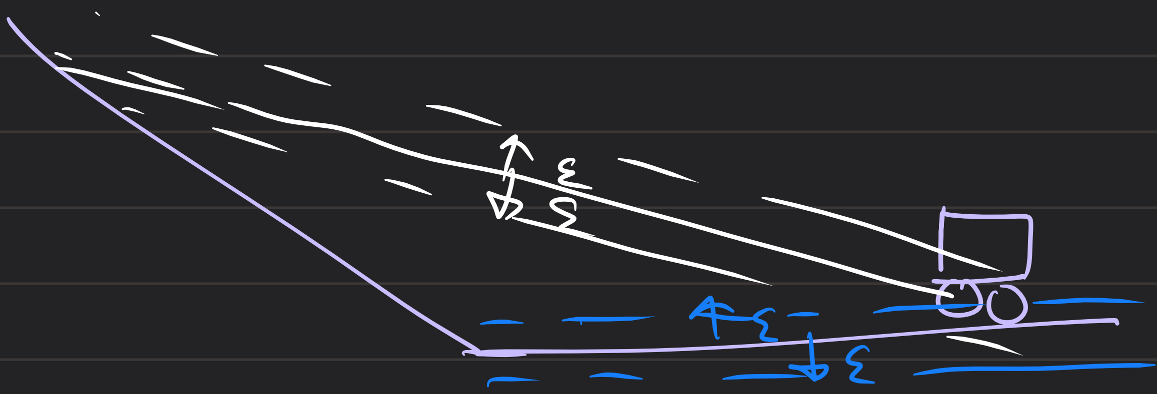
m-1: mark a z-coord & remove all the points below it

→ but this is not a good method becz → when car tilts downwards or what about a curve

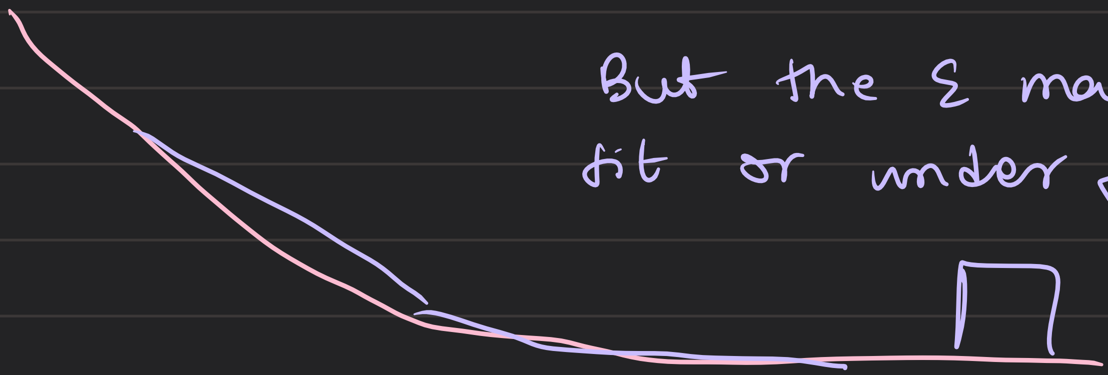
→ we use ransac

what is ransac, how it works?

it randomly choose 3 points & make a plane detects all the points which are in the error bound (ε) of the plane is choosen, checks many planes, the plane which has the max. no. of points is choosen as a ground plane & all the points in the ε of the plane.

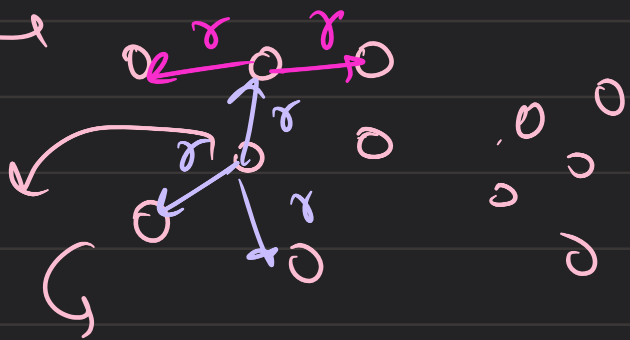


But the ε may over fit or under fit.



Clustering ?

using DBSCAN →



core point

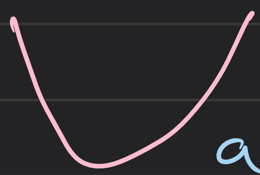
connected to Boundary points

Boundary point

↳ cannot extend further

Classification of cones: blue-white / yellow-black.

blue-white

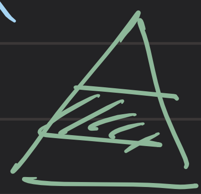


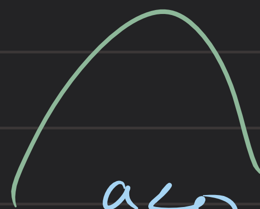
$a > 0$

High
Low
High

yellow-black

Low
High
Low

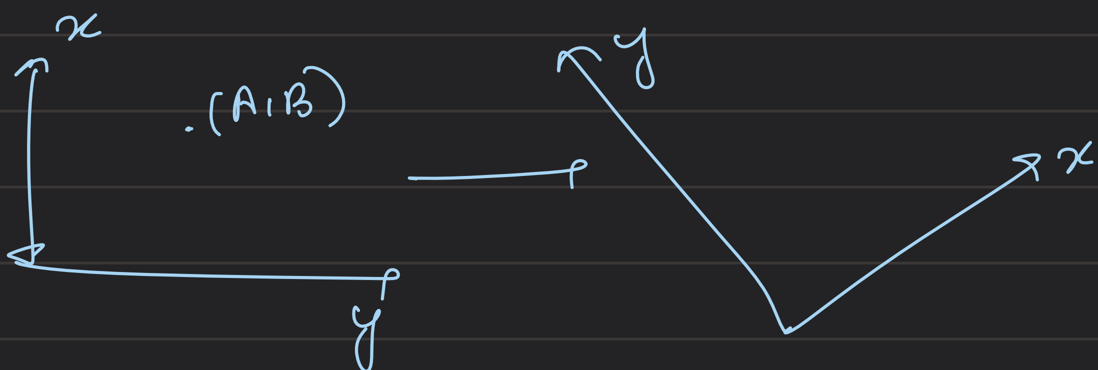




$a < 0$

Basically fixing a parabola

Meet →



shift & rotate
(translate)

$$\begin{bmatrix} A \\ B \end{bmatrix} + \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_{new} \\ y_{new} \end{bmatrix}$$

translation

rotation.

angle to be changed → yaw

Lidar → Camera (3D → 2D)

for 3D

$$\begin{bmatrix} A \\ B \end{bmatrix} + \begin{bmatrix} M \end{bmatrix} \begin{bmatrix} x_{new} \\ y_{new} \end{bmatrix}$$

focal length

↳ when we zoom using a camera → depth changes

↳ problem for LIDAR

Lidar → Camera → Output

(change the frame)

from both camera & LiDAR

$\xrightarrow{\text{yolo}}$ (x,y,z) using lidar & colour using camera